

ECO7101

Handout for Mid2

1. Explain utility maximization of a consumer in terms law of equi-marginal utility or equating marginal utilities per dollar.

According to the law of equi-marginal utility or equating marginal utilities per dollar the equilibrium that occurs when the consumer has spent all of his or her income and the marginal utilities per dollar spent on each good purchased are equal:

$$\frac{MU_O}{P_O} = \frac{MU_A}{P_A} = \dots\dots\dots$$

Where O, A.... are goods.

Suppose there are only two goods in the world: apples and oranges. At present, a consumer is spending his entire income consuming 10 apples and 10 oranges a week. For a particular week, the marginal utility (MU) and price (P) of each are as follows:

$$MU_{\text{oranges}} = 30 \text{ utils}$$

$$MU_{\text{apples}} = 20 \text{ utils}$$

$$P_{\text{oranges}} = \$1$$

$$P_{\text{apples}} = \$1$$

So, the consumer's marginal (last) dollar spent on apples returns 20 utils per dollar, and his marginal (last) dollar spent on oranges returns 30 utils per dollar. Here,

$$\frac{MU_O}{P_O} = \frac{30}{1} = 30$$

$$\frac{MU_A}{P_A} = \frac{20}{1} = 20$$

The ratio MU_O / P_O (O = oranges) is greater than the ratio MU_A/P_A (A = apples).

$$\text{Therefore } \frac{MU_O}{P_O} > \frac{MU_A}{P_A}$$

Since oranges give more utility the consumer will buy more oranges and will buy less number of apples as apples give less utility. As a result number of oranges increase that reduces MU_O and as number of apples decreases that increases MU_A . If the process continues MU_O and MU_A will be equal at a point in time. It confirms consumer's equilibrium. The equilibrium that occurs when the consumer has spent all of his or her income and the marginal utilities per dollar spent on each good purchased are equal:

$$\frac{MU_O}{P_O} = \frac{MU_A}{P_A} = \dots\dots\dots$$

A person in consumer equilibrium has maximized total utility. By spending his or her money on goods that give the greatest marginal utility and, in the process, bringing about the consumer equilibrium condition, the consumer is adding as much to total utility as is possible.

2. Explain the properties of an indifference curve.

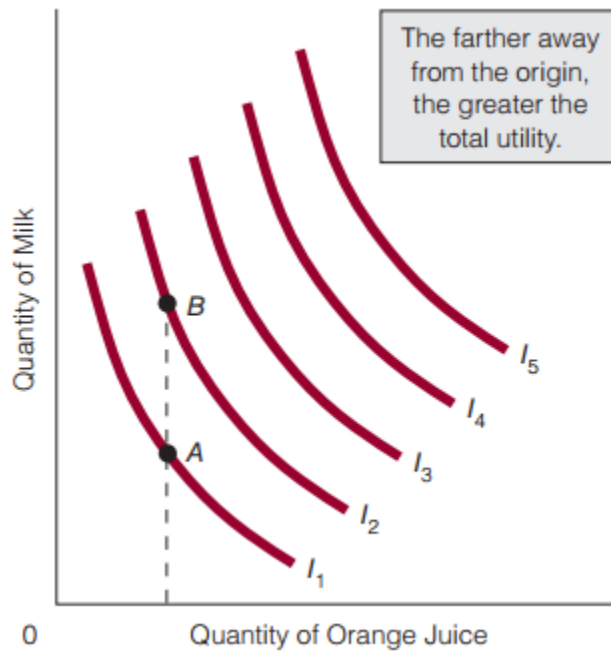
(a) Indifference curves are downward sloping (from left to right).

The assumption that consumers always prefer more of a good to less requires that indifference curves slope downward from left to right. More simply, indifference curves are downward sloping because a person has to get more of one good in order to maintain the same level of satisfaction (utility) when giving up some of another good.

(b) Indifference curves are convex to the origin. As we move down to the right along the indifference curve, it becomes flatter. The absolute value of the slope of the indifference curve is the marginal rate of substitution (MRS). The MRS is the amount of one good an individual is willing to give up to obtain an additional unit of another good to maintain equal total utility.

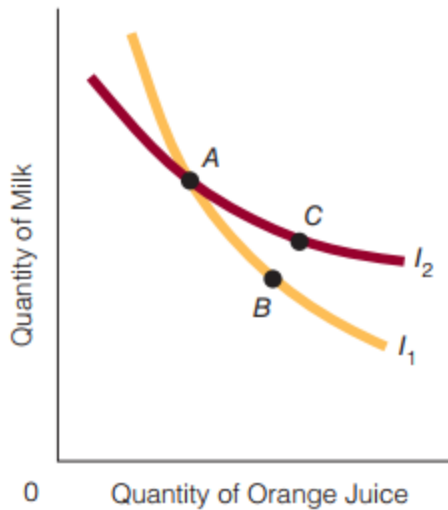
Since MRS is negative or MRS declines that's why an IC is convex to the origin.

(c) Higher IC gives higher utility



Higher IC contains more goods. More goods give more utility. Indifference curves farther away from the origin represent greater total utility than those closer to the origin.

(d) Two ICs never intersect



Suppose two ICs, I_1 and I_2 (where I_1 is lower IC and I_2 is higher IC) intersect at A. It implies that the higher IC, I_2 and lower IC, I_1 give same utility at A. It cannot be as it violates transitivity assumption or violates property 3 (Higher IC gives higher utility).

3. Explain the budget equation. When does budget line rotate and shift?

Budget line or equation represent all the combinations, or bundles, of two goods a person can purchase, given a certain money income and prices for the two goods.

Budget Constraint can also be expressed by the following equation known as budget equation.

$$M = P_1X_1 + P_2X_2$$

M = money income

X_1 = commodity 1

$$M = P_1X_1 + P_2X_2$$

P_1 = Price of commodity 1

Lets solve it for X_2

X_2 =Commodity 2

$$\Rightarrow P_2X_2 = M - P_1X_1$$

P_2 = Price of x_2

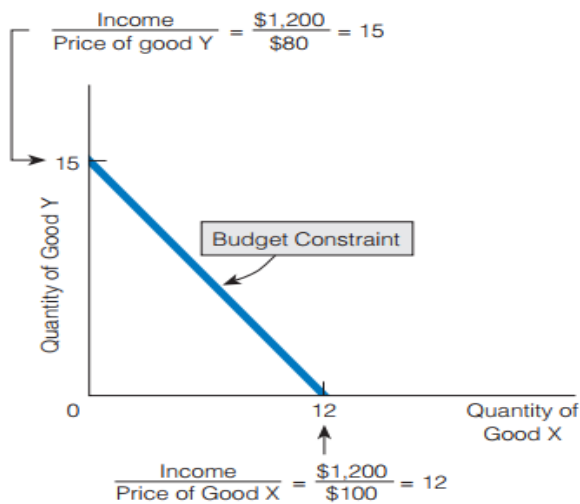
$$\Rightarrow X_2 = \frac{M}{P_2} - \frac{P_1}{P_2}x_1$$

P_1X_1 = expenditure on X_1

P_2X_2 - expenditure on X_2

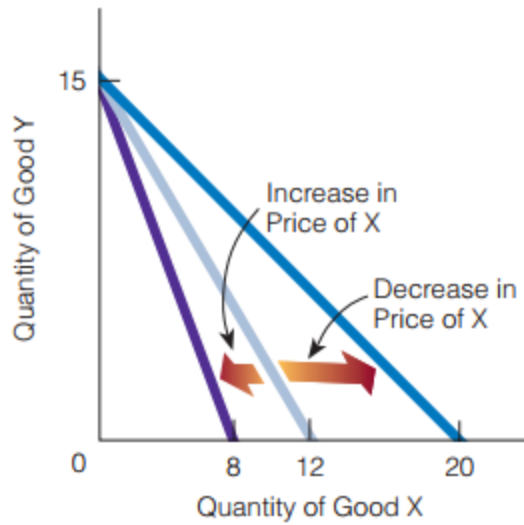
$\frac{M}{P_2}$ = Purchasing power or real income

$\frac{p_1}{p_2}$ = relative price or slope of the budget line

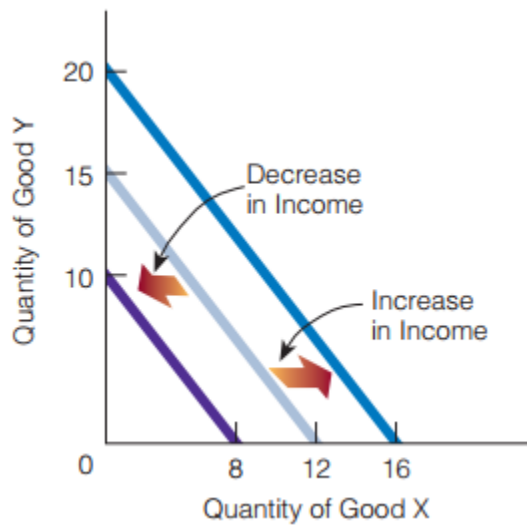


Suppose income of a consumer is Tk. 1200. Suppose price of X_1 commodity is Tk. 100 and price X_2 is Tk. 80. If the consumer spend all his money on either X_1 or X_2 , consumer can purchase either 15 units of X_2 or 12 units of X_1 . By combining these two extreme points we can derive budget line.

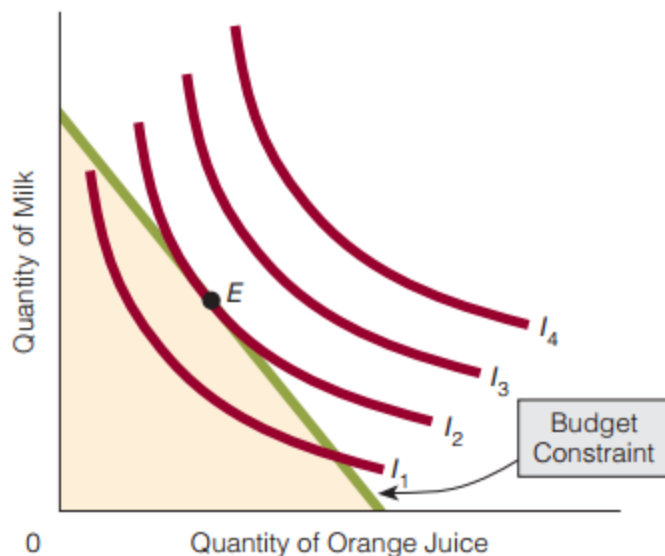
Budget line rotates if prices change or relative price (P_1/P_2) changes or the slope of the budget line changes but real income remains unchanged.



Budget line shifts if real income changes but relative price (P_1/P_2) or the slope of the budget remains unchanged.



4. Explain equilibrium or utility maximization of a consumer in terms indifference curve analysis.



Consumer equilibrium exists at the point where the slope of the budget constraint is equal to the slope of an indifference curve or where the budget constraint is tangent to an indifference curve. In the exhibit, this point is E. Here, $\frac{P_o}{P_m} = \frac{MU_o}{MU_m}$.

The absolute value of the slope of the indifference curve at any point is the marginal rate of substitution, which is equal to the marginal utility of one good divided by the marginal utility of another good, or $\frac{MU_o}{MU_m}$.

The necessary condition for consumer equilibrium is obviously that the individual will try to reach a point on the highest indifference curve possible. This point is where the slope of the budget constraint is equal to the slope of an indifference curve (or where the budget constraint is tangent to an indifference curve). At that point, consumer equilibrium is established and the following condition holds:

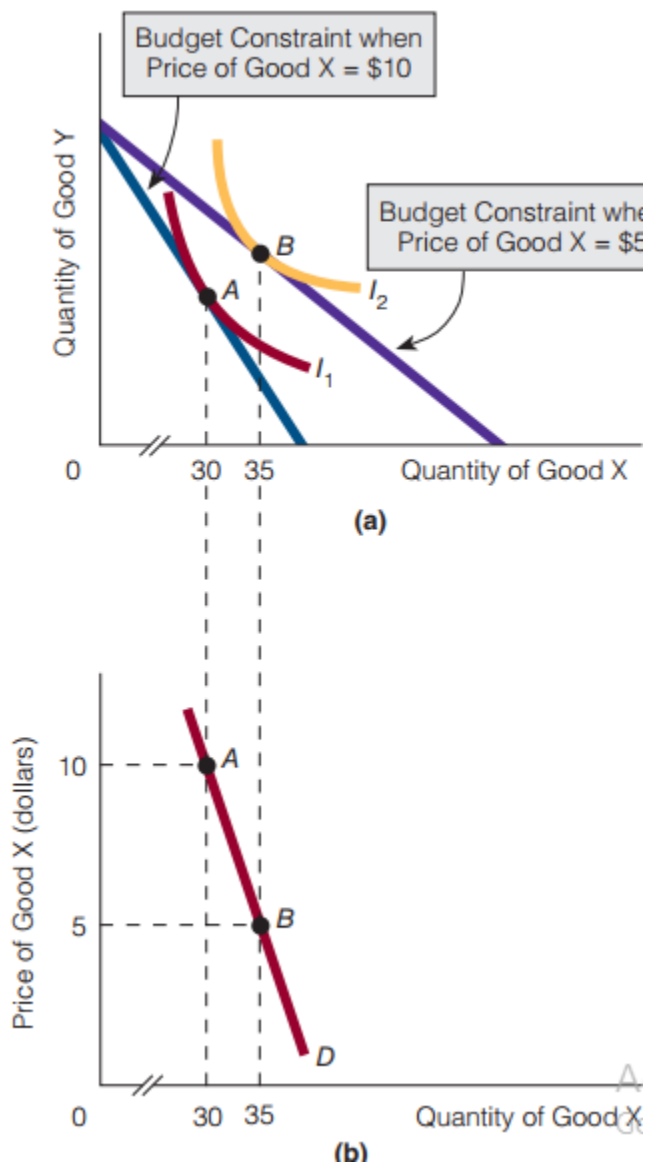
$$\frac{P_o}{P_m} = \frac{MU_o}{MU_m}$$

In the figure, the preceding condition is met at point E.

5. Derive demand curve from indifference curve analysis.

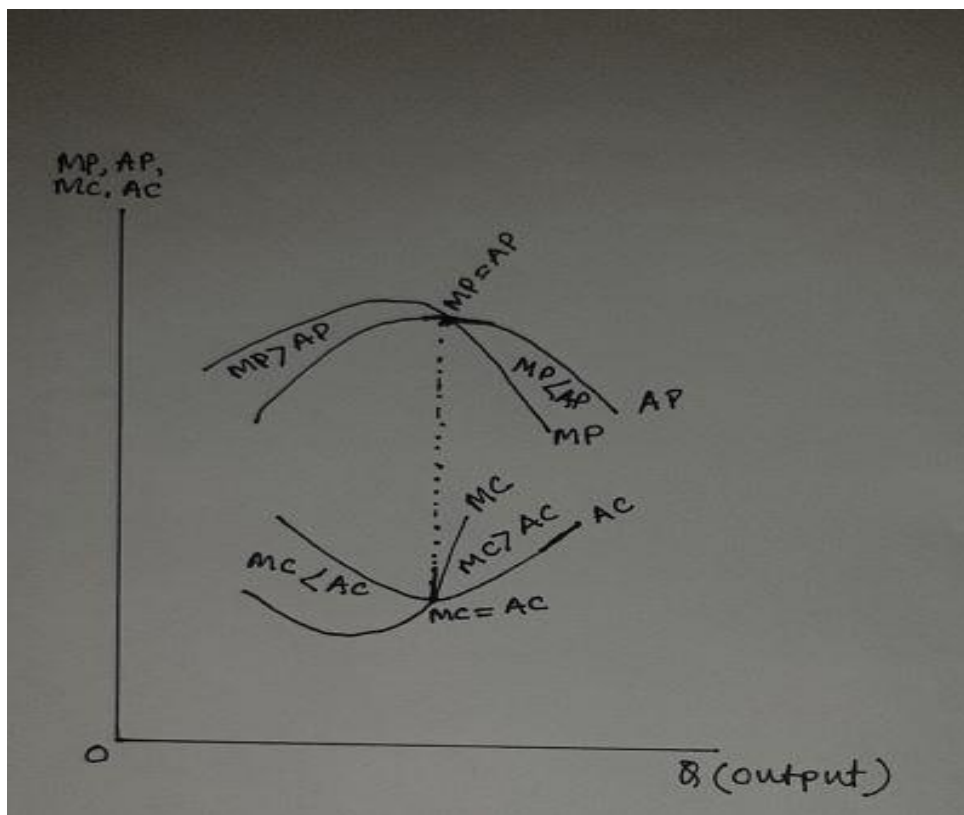
We can now derive a demand curve within a budget constraint–indifference curve framework.

The following figure shows two budget constraints, one reflecting a \$10 price for good X and the other reflecting a \$5 price for good X. As the price of X falls, the consumer moves from point A to point B. At B, 35 units of X are consumed; at A, 30 units of X are consumed. So, a lower price for X results in greater consumption of X. By plotting the relevant price and quantity data, we derive a demand curve for good X in the following figure.



From Indifference Curves to a Demand Curve (a) At a price of \$10 for good X, consumer equilibrium is at point A, with the individual consuming 30 units of X. As the price falls to \$5, the budget constraint moves outward (away from the origin) and the consumer moves to point B and consumes 35 units of X. Plotting the price–quantity data for X gives a demand curve for X in

6. Explain the relationship between MP and AP; MC and AC and MP-AP and MC-AC



As long as MP is rising AP is also rising and $MP > AP$. At the highest point of AP, MP intersects AP. At the intersection point $MP = AP$. Beyond that as MP falls AP also falls and $MP < AP$.

As long as MC is falling AC is also falling and $MC < AC$. At the lowest point of AC, MC intersects AC. At this point $MC = AC$. Beyond that as MC rises and AC also rises and $MC > AC$.

When MP and AP are rising, MC and AC are falling. When $MP = AP$, $MC = AC$. Beyond that if MP and AP fall, MC and AC rise.

7. Explain the properties of a perfect competitive

(a) There are many sellers and many buyers, none of which is large in relation to total sales or purchases.

Given many buyers and sellers, each buyer and each seller may act independently of other buyers and sellers, respectively, and each is such a small part of the market as to have no influence on price, they are price takers.

(b) Each firm produces and sells a homogeneous product.

Each firm sells a product that is indistinguishable from all other firms' products in a given industry. As a consequence buyers are indifferent to the sellers.

(c) Buyers and sellers have all relevant information about prices, product quality, sources of supply, and so forth.

Buyers and sellers know who is selling what, at what prices, at what quality, and on what terms. In short, they know everything that relates to buying, producing, and selling the product.

(d) Firms have easy entry and exit.

New firms can enter the market easily, and existing firms can exit the market easily. There are no barriers to entry or exit.

8. Explain the profit maximization of a short-run competitive firm or what Level of output does the profit-maximizing firm produces in the short-run?

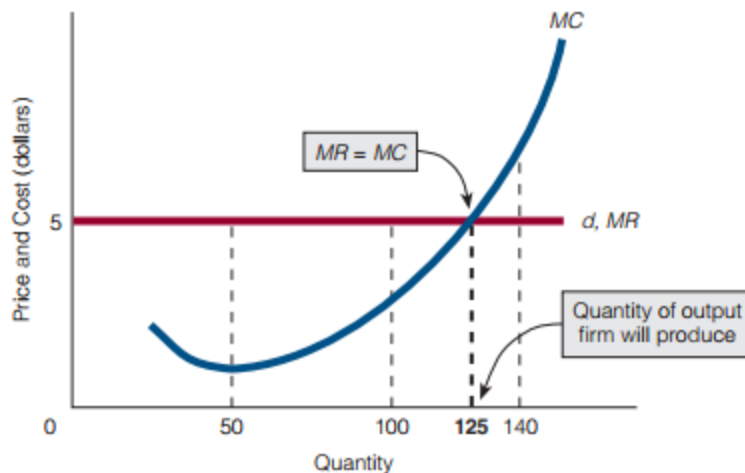
A competitive firm is a price taker as it takes the price determined by market demand and supply. Therefore the firm decides how much to produce to maximize profit. **Profit Maximization Rule** tells us that profit is maximized by producing the quantity of output at which $MR = MC$.

In Exhibit 3, the perfectly competitive firm's demand curve (d) and marginal revenue curve (MR, which is the same as d) are drawn at the equilibrium price of \$5. The firm's marginal cost curve (MC) is also shown. On the basis of these curves, the firm will continue to increase its quantity of output as long as marginal revenue is greater than marginal cost. It will not produce units of output for which marginal revenue is less than marginal cost. Therefore, the firm will stop increasing its quantity of output when marginal revenue and marginal cost are equal. The profit maximization rule for a firm says, produce the quantity of output at which $MR = MC$. In Exhibit 3, $MR = MC$ at price 5 produces 125 units of output. For the perfectly competitive firm, the profit maximization rule can be written as $P = MC$ because, for that firm, $P = MR$ and $MR = MC$. Therefore, in perfect competition, profit is maximized when, $P = MR = MC$.

EXHIBIT 3

The Quantity of Output That the Perfectly Competitive Firm Will Produce

The firm's demand curve is horizontal at the equilibrium price. Its demand curve is its marginal revenue curve. The firm produces that quantity of output at which $MR = MC$.



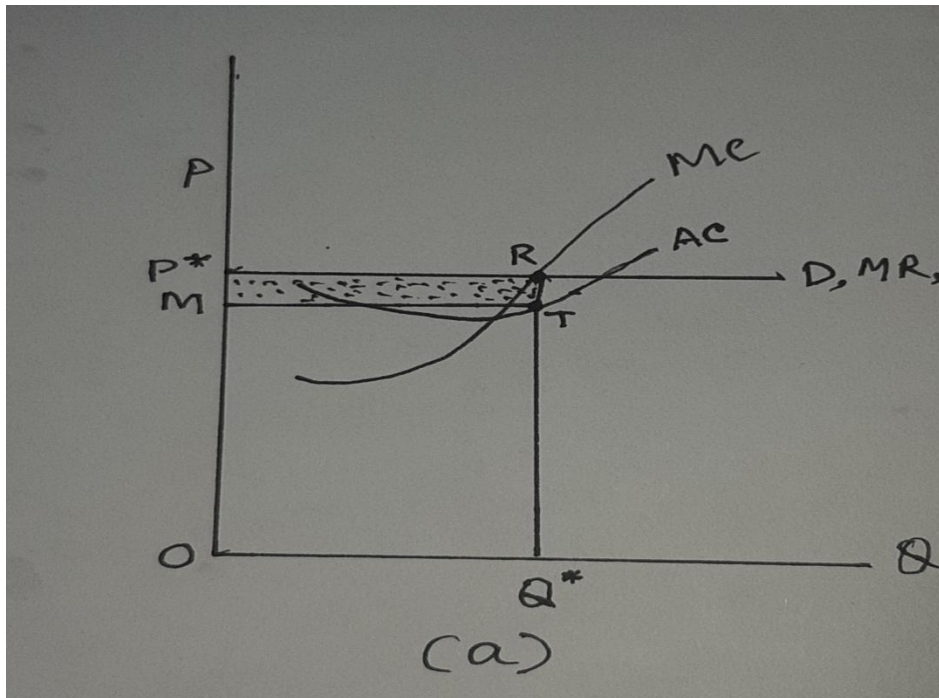
On the basis of above diagram, the firm maximizes profit at a price \$5 and quantity 125 units for which $P = MR = MC$.

9. What are the short-run possible outcomes of a competitive firm? Explain.

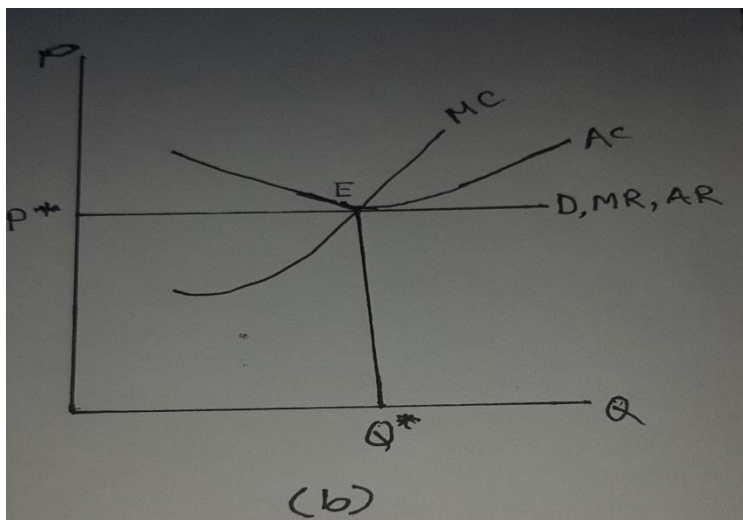
In the short-run a competitive firm has three possible outcomes:

(1) Firm can make economic profit

Perfect competitive firm in the short-run may make economic profit in terms per unit profit or total profit. At point R, $MR = MC$. Competitive firm produces Q^* amount of output and sells at market determined price P^* . Average revenue, $AR = Q^*R$ and average cost, $AC = Q^*T$. Therefore, $AR > AC$. The firm makes economic profit per unit is RT . Total economic profit for Q^* amount of output is $MP \cdot RT$.



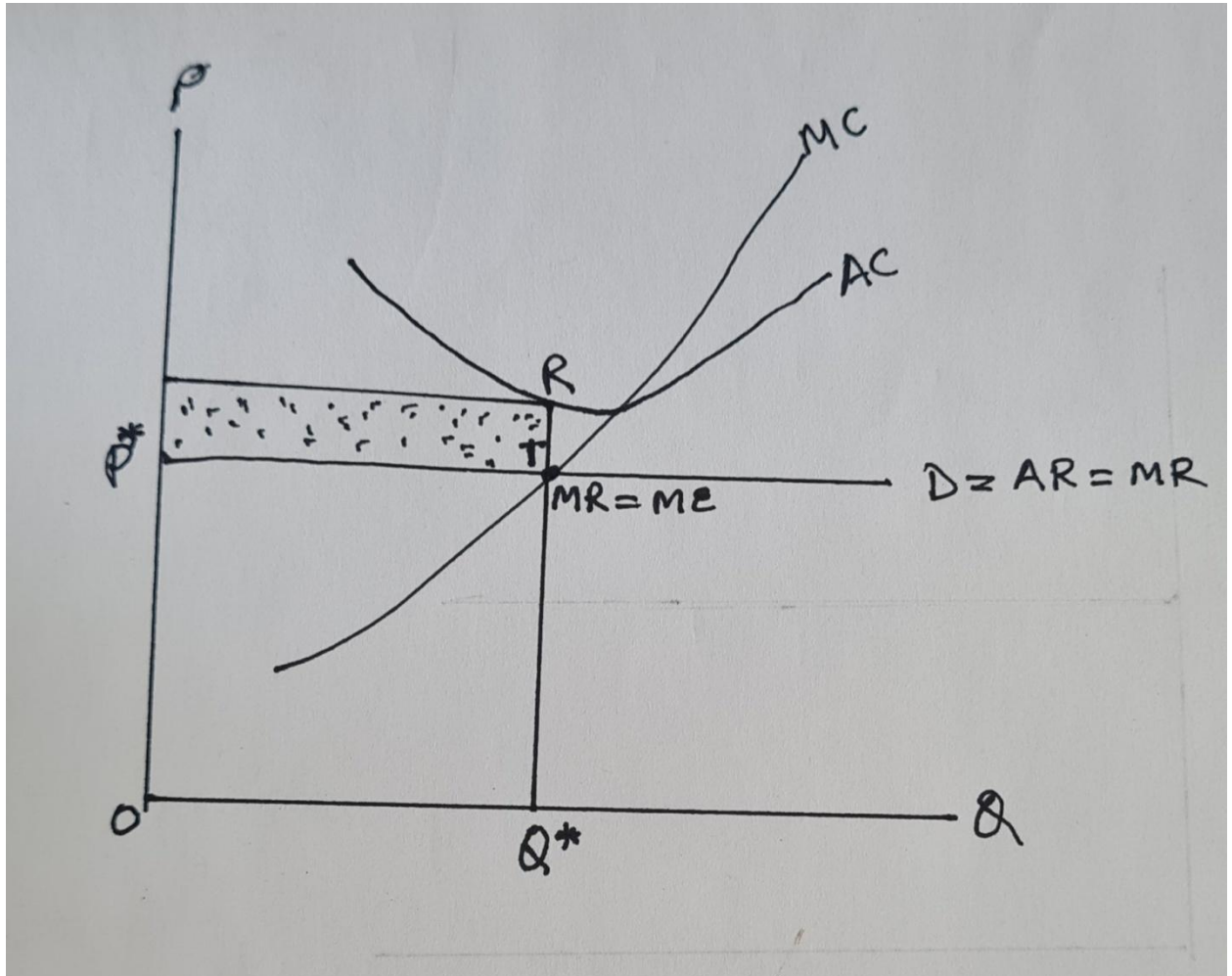
(2) Can make normal profit



Short-run competitive firm makes normal or zero economic profit if $TR = TC$ or $AR = AC$. In figure (b) competitive firm produces Q^* amount of output and sells at market determined price P^* . At equilibrium E, $MR = MC$ and also $AR = AC$. So that in this situation short-run competitive firm can make only normal or zero economic profit.

(3) Can incur economic loss

Short-run competitive firm may incur economic loss. Economic loss occurs if $TR - TC < 0$ or $TR < TC$ or $TC > TR$ or $AR < AC$ or $AC > AR$. In the following figure: $Q^*R = AC$, $Q^*T = AR$. $AC > AR$ by the amount RT which is per unit economic loss. Total economic loss = $Q^* \times RT = P^*MRT$.



PERFECT COMPETITION IN THE LONG RUN

The number of firms in a perfectly competitive market may not be the same in the short run as in the long run. For example, if the typical firm is making economic profits in the short run, new firms will be attracted to the industry and the number of firms will increase. If the typical firm is sustaining losses, some existing firms will exit the industry and the number of firms will decrease.

The following conditions characterize long-run competitive equilibrium:

1. Economic profit is zero; that is, price (P) is equal to short-run average total cost (SRATC): $P = \text{SRATC}$. If firms in an industry make positive economic profits in the short-run some new firms will enter the industry. As a result supply will increase and price will decrease. If some existing firms make economic loss in the short-run some will quit the industry in the long-run. As a result supply will decrease and price will increase. Long-run competitive equilibrium cannot exist if firms have an incentive to enter or exit the industry in response to positive economic profits or losses. For long-run equilibrium to exist, there can be no incentive for firms to enter or exit. This condition is brought about by zero economic profit (normal profit), which is a consequence of the equilibrium price being equal to short-run average total cost.
2. Firms are producing the quantity of output at which price (P) is equal to marginal cost (MC):

$$P = MC$$

Perfectly competitive firms naturally move toward the output level at which marginal revenue (or price, because, for a perfectly competitive firm, $MR = P$) equals marginal cost.

3. No firm has an incentive to change its plant size to produce its current output; that is, at the quantity of output at which $P = MC$

The three conditions necessary for long-run competitive equilibrium can be stated: long-run competitive equilibrium exists when $P = MC = \text{SRATC} = \text{LRATC}$. Long-run competitive equilibrium exists when firms have no incentive to make any changes—that is, when there is no incentive for firms to do any of the following:

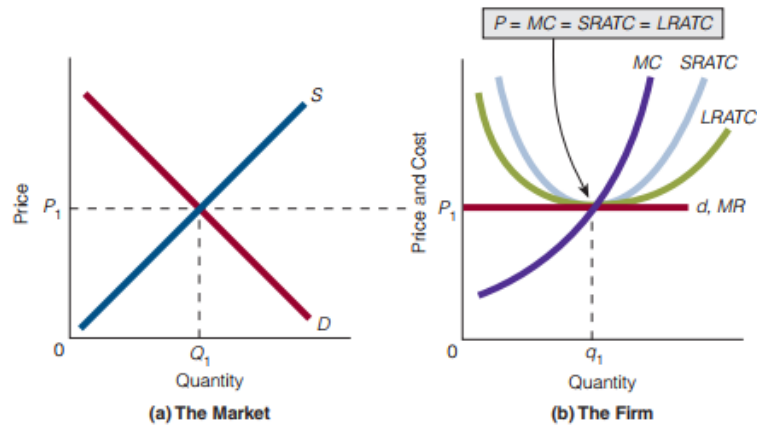
1. enter or exit the industry.
2. produce more or less output.

3. change their plant size.

EXHIBIT 9

Long-Run Competitive Equilibrium

(a) Equilibrium in the market.
(b) Equilibrium for the firm. In (b), $P = MC$ (the firm has no incentive to move away from the quantity q_1 of output at which this equality occurs), $P = SRATC$ (there is no incentive for firms to enter or exit the industry), and $SRATC = LRATC$ (there is no incentive for the firm to change its plant size).



A firm that produces its output at the lowest possible per-unit cost (lowest ATC) is said to exhibit productive efficiency. The perfectly competitive firm is productively efficient in long-run equilibrium.